

COMP3311

Week 2 Tutorial



Agenda

-
- ✓ Admin Stuff
-
- ✓ Week 1 key concepts
-
- ✓ Tutorial questions
-
- ✓ Demo: how to ssh into vxdb02
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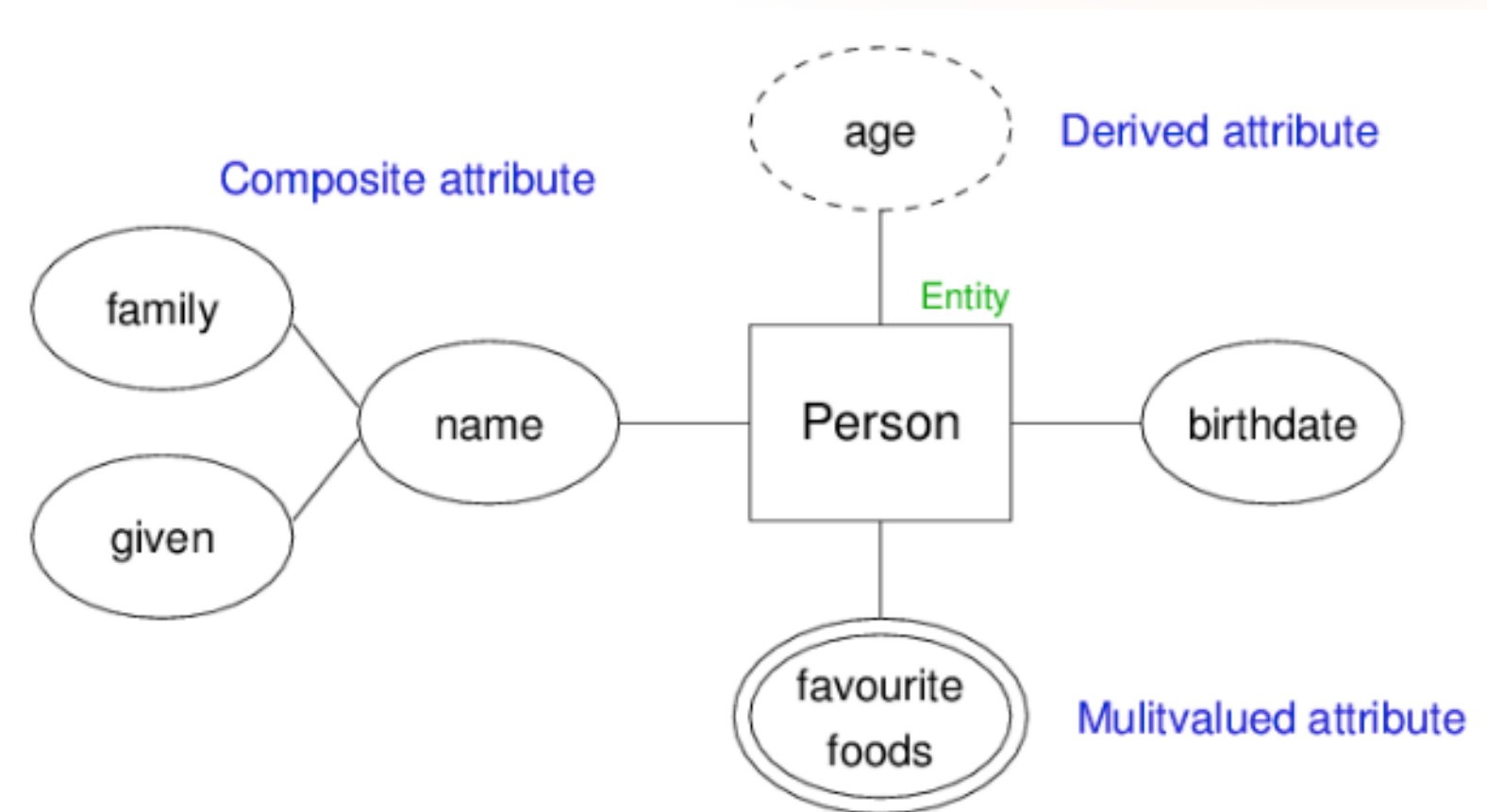
Admin Stuff

- **Getting help:**
 - Discourse (forum)
 - Help sessions (for assignments)
 - Class email (cs3311@cse.unsw.edu.au)
 - Email me (z5479432@ad.unsw.edu.au)
- **Assessment Structure**
 - Quizzes (Weeks 2,3,4,7,8,10) - 12%
 - Assignment 1 (Week 5) - 13%
 - Assignment 2 (Week 10) - 15%
 - Final (Exam period) - 60%
- **Tutorial**
 - 1.5 hr class going through tutorial questions
 - Tutorial questions are based on content from the week before
 - I like to go through the core concepts you'll need to use to answer the questions so please feel free to come along even if you're not caught up to lectures :D
 - Tutorial questions help you prepare for assignments and especially the final exam
 - Solutions for tutorial questions come out the week after
 - To access the work done in these tutorials:
<https://github.com/nicoleewang/tutoring/tree/main/3311/26t2/H10A>

Week 1 Key Concepts

Data Modelling - ER Diagram

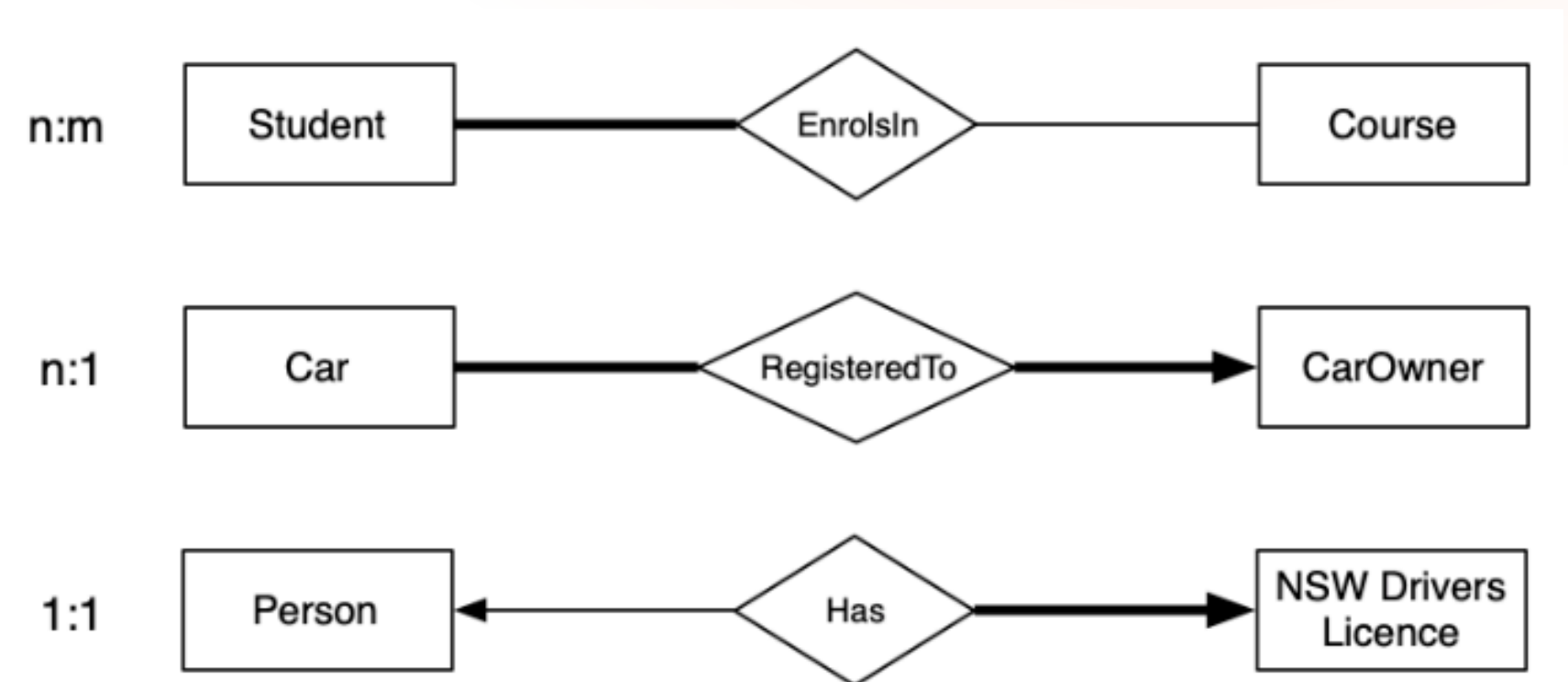
- ER diagram: modelling notation containing collections of
 - entities (**rectangles**) and their associated attributes (**ovals**)
 - relationships (**diamonds**) between entities
- Special types of attributes:



Week 1 Key Concepts

Data Modelling - ER Diagram

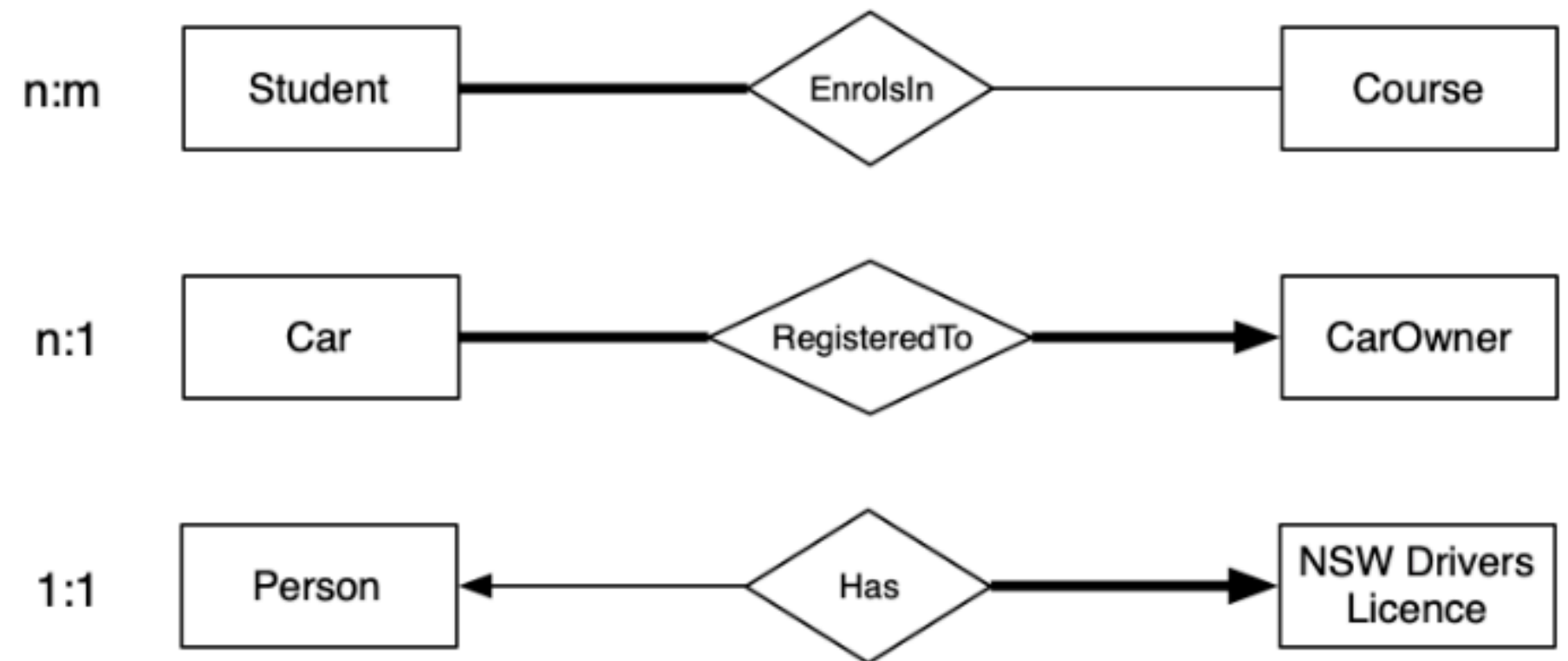
- Relationships
 - Thick/Thin lines represent **participation**
 - Thick line = total participation
 - e.g. (diagram 1) to be a Student you NEED to be enrolled in a course
 - Thin line = partial participation
 - e.g. (diagram 1) a Course can have Students enrolled in it (but not must)
 - How to read the diagram
 - The line next to the entity describes the participation of that entity



Week 1 Key Concepts

Data Modelling - ER Diagram

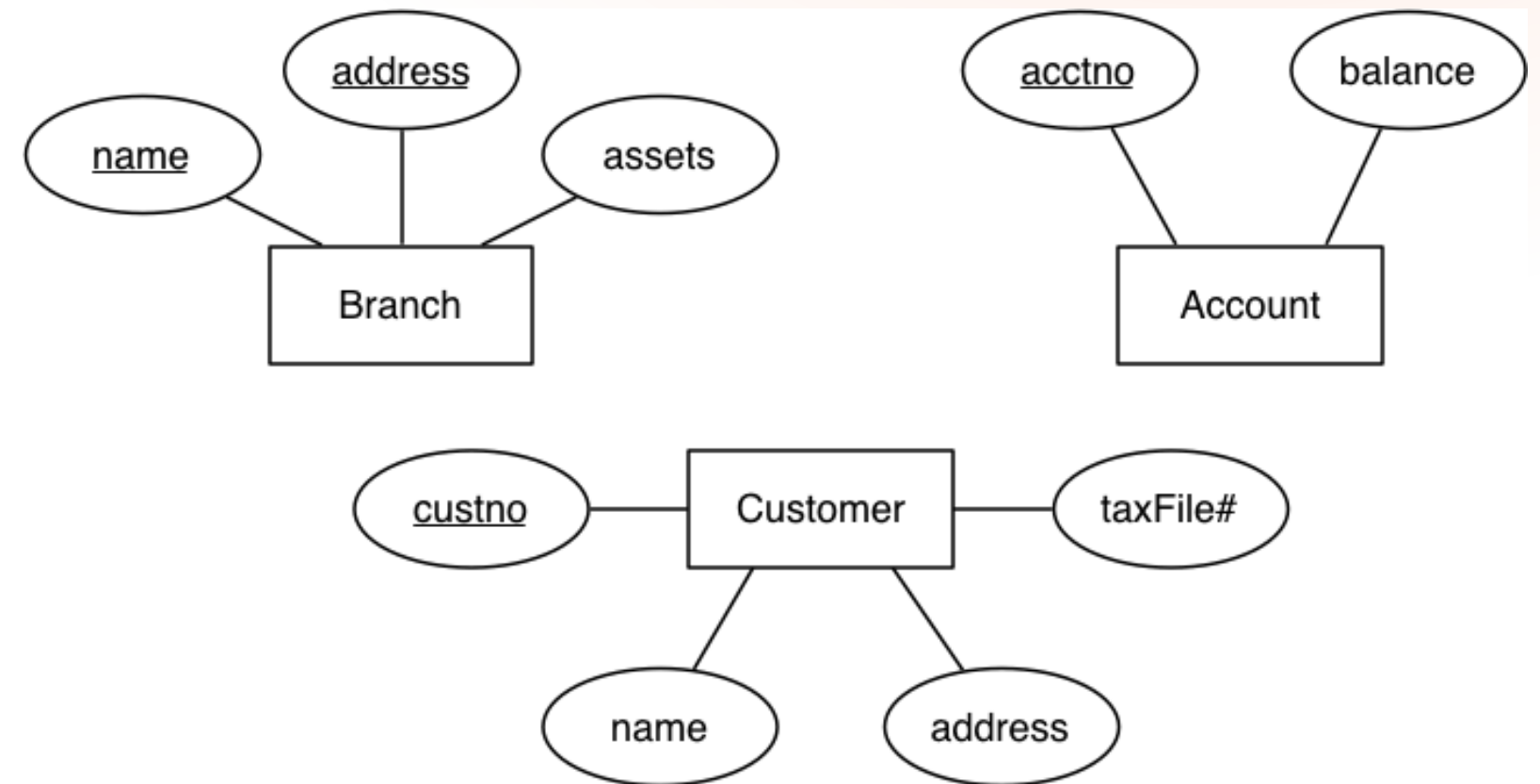
- Relationships
 - Arrows indicate **cardinality**
 - No arrow = any number
 - e.g. (diagram 1) a Student can enrol in as many Courses as they want and a Course can have as many students enrolled in it
 - Arrow = at most 1
 - e.g. (diagram 3) a Person can have at most one NSW Driver Licence
 - How to read the diagram:
 - An arrow touching an entity means that entity participates at most once in the relationship



Week 1 Key Concepts

Data Modelling - ER Diagram

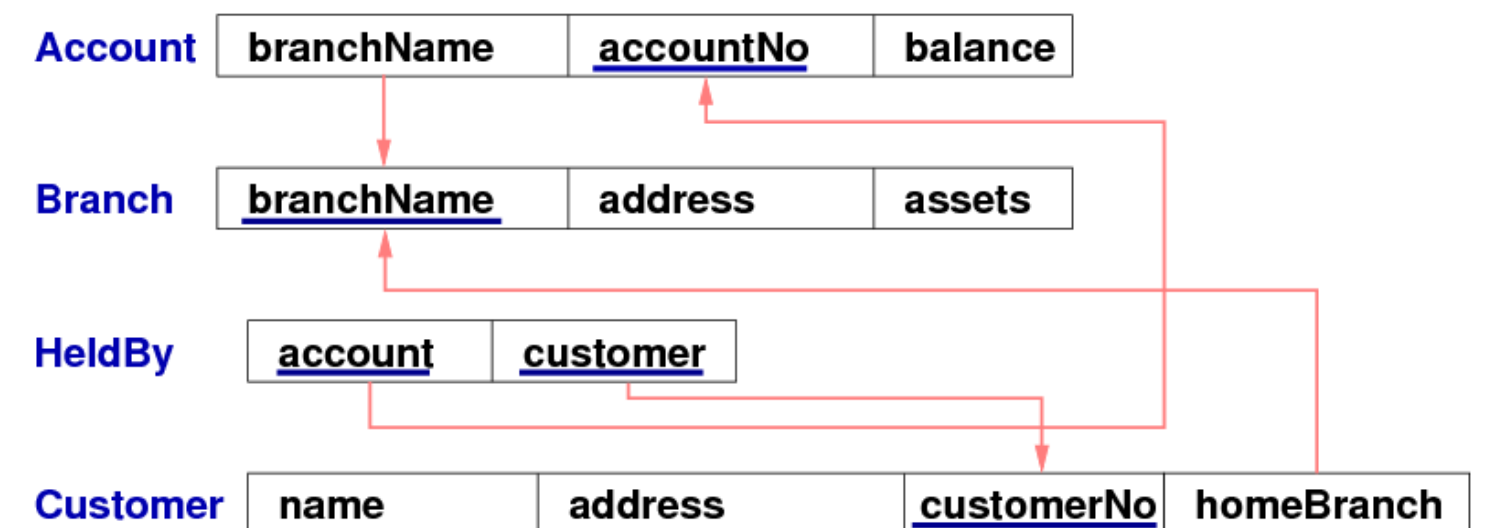
- Primary Keys
 - Entities have attribute(s) that are used to identify entity instances from one another
 - primary keys are NOT NULL and UNIQUE for all instances in the set
 - in an ER diagram: primary keys are indicated by an underline
 - keys with more than one attribute are called composite keys



Week 1 Key Concepts

Data Modelling - Relational Model

- What is it?
 - Relational modelling is a different type of modelling compared to ER diagrams
 - models data as a collection of relations (tables/set of tuples) with attributes
 - Better represents the structure of the schema when converting to SQL
- What does it look like?
 - Relation Schemas (usually denoted R,S,T,...) represent the structure of the relation, and has:
 - A name (unique within the database)
 - A set of attributes (can be seen as column headings for a table)
 - e.g. Account(branchName, accountNo, balance)



Tutorial Questions

Now let's try some questions :D

- Q1
- Q7
- Q9
- Q11
- Q15
- Q18

Demo: how to ssh into vxdb02

Setting Up

- Follow [prac exercise 01](#) for detailed instructions on how to set up!
- How to set up PostgreSQL:
 - install on your home computer
 - set it up on your CSE account and access through VLab or SSH (I recommend this)
- Setup on your CSE account:
 - from cse servers, ssh via:
 - `ssh vxdb02`
 - from home, ssh via:
 - `ssh YourZID@vxdb02.cse.unsw.edu.au`
 - once you ssh run this to start the server (need to do this every time)
 - `source /localstorage/$USER/env`
 - `p1`
- Ending your session
 - please ensure you run the following when you are done
 - `p0`

Demo: how to ssh into vxdb02

Useful Commands

- Within the PostgreSQL server (in a linux shell):
 - `psql -l`
 - to see what databases you have
 - `createdb <db name>`
 - create a new db
 - `psql <db name>`
 - access the database
 - `psql <db name> -f <file name>`
 - to populate your database with a SQL file
 - `p1`
 - start server
 - `p0`
 - stop server
- Inside psql (within a database):
 - `\d`
 - to find relations within your database
 - `\d <table name>`
 - to find more details about an individual table
 - `\q`
 - quit
 - other SQL queries...

Reminders for the Week

Have you done these things???

- ☒ Quiz 1 (Due this Friday 11:59pm)
- ☒ Try to ssh into vxdb02 on your machine, run the server and play around with it